VidyaSetu

Student Writing and Deadline Companion | Project Summary
Built by Samarth Mishra | Version 1.0.0 | October 26, 2026

1. Purpose

VidyaSetu is a full-stack academic integrity and deadline companion for students. It compares a draft against reference sources the student provides, formats citations in APA 7, MLA 9, and IEEE styles, converts a word count and due date into a day-by-day assignment plan, and hosts a study exchange board for peer feedback requests.
The application follows the same engineering values as the MediSetu reference project: static frontend plus Node.js serverless REST APIs, zero accounts, zero trackers, zero external API keys, full privacy and terms pages, and a strict automated test suite.

2. Architecture

Frontend: HTML5, CSS3, and ES6+ JavaScript with a flat zinc and emerald design system, a live API response inspector drawer, and audio read-aloud via the browser speech engine.
Backend: Five self-contained serverless endpoints served by Node.js (local server.js for development, Vercel serverless functions in production): GET /api/health, POST /api/check, POST /api/cite, POST /api/plan, and GET/POST /api/board.
Deployment: Vercel with clean URLs and an SVG favicon rewrite. No databases, no third-party services, and no environment secrets are required.

3. Originality Engine

Draft and source text are normalized, split into sentences, and compared with word-level containment plus 5-word shingle containment. Per-source match percentages feed an overall score that blends the strongest source (70 percent) with the average (30 percent). Sentence pairs above threshold are flagged as verbatim, high, or review severity and shown with both excerpts side by side.

4. Citation Builder and Planner

The citation formatter covers website, journal article, book, and video sources across APA, MLA, and IEEE styles, with per-style in-text citation hints. The assignment planner estimates effort from word count, work type, and daily hours, then emits a day-by-day schedule with pacing feedback and three standard checkpoints, persisted only in browser localStorage.

5. Testing and Standards

test_suite.mjs runs a static standards audit over all 15 project files (zero em dashes, zero emoji, zero purple palette, zero gradients, zero pill radii, zero stock images, zero forbidden AI tags, favicon and legal links on every page, full DOM ID parity) followed by three end-to-end HTTP passes covering every route and API. All checks pass with zero violations.

6. Hackathon Fit (IceHacks 2026)

Judging criteria coverage: Technical Execution (5 working serverless endpoints plus a 3-pass test suite), Creativity and Originality (a self-comparison originality engine instead of a generic to-do app), Impact and Usefulness (directly useful to any student writer before a deadline), Design and User Experience (accessible flat design, keyboard-friendly controls, audio read-aloud), Presentation (this summary plus the live deployed demo).